Overview of the Coupled Model Intercomparison Project Phase 6 (CMIP6) experimental design and organization

Authors: Veronika Eyring, Sandrine Bony, Gerald A. Meehl, Catherine A. Senior, Bjorn Stevens, Ronald J. Stouffer, Karl E. Taylor

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Abstract

By coordinating the design and distribution of global climate model simulations of the past, current, and future climate, the Coupled Model Intercomparison Project (CMIP) has become one of the foundational elements of climate science. However, the need to address an ever-expanding range of scientific questions arising from more and more research communities has made it necessary to revise the organization of CMIP. After a long and wide community consultation, a new and more federated structure has been put in place. It consists of three major elements: (1) a handful of common experiments, the DECK (Diagnostic, Evaluation and Characterization of Klima) and CMIP historical simulations (1850–near present) that will maintain continuity and help document basic characteristics of models across different phases of CMIP; (2) common standards, coordination, infrastructure, and documentation that will facilitate the distribution of model outputs and the characterization of the model ensemble; and (3) an ensemble of CMIP-Endorsed Model Intercomparison Projects (MIPs) that will be specific to a particular phase of CMIP (now CMIP6) and that will build on the DECK and CMIP historical simulations to address a large range of specific questions and fill the scientific gaps of the previous CMIP phases. The DECK and CMIP historical simulations, together with the use of CMIP data standards, will be the entry cards for models participating in CMIP. Participation in CMIP6-Endorsed MIPs by individual modelling groups will be at their own discretion and will depend on their scientific interests and priorities. With the Grand Science Challenges of the World Climate Research Programme (WCRP) as its scientific backdrop, CMIP6 will address three broad questions: How does the Earth system respond to forcing?What are the origins and consequences of systematic model biases?How can we assess future climate changes given internal climate variability, predictability, and uncertainties in scenarios? This CMIP6 overview paper presents the background and rationale for the new structure of CMIP, provides a detailed description of the DECK and CMIP6 historical simulations, and includes a brief introduction to the 21 CMIP6-Endorsed MIPs.

Introduction

The Coupled Model Intercomparison Project (CMIP) organized under the auspices of the World

Climate Research Programme's (WCRP) Working Group on Coupled Modelling (WGCM) started 20 years ago as a comparison of a handful of early global coupled climate models performing experiments using atmosphere models coupled to a dynamic ocean, a simple land surface, and thermodynamic sea ice (Meehl et al., 1997). It has since evolved over five phases into a major international multi-model research activity (Meehl et al., 2000, 2007; Taylor et al., 2012) that has not only introduced a new era to climate science research but has also become a central element of national and international assessments of climate change (e.g. IPCC, 2013). An important part of CMIP is to make the multi-model output publicly available in a standardized format for analysis by the wider climate community and users. The standardization of the model output in a specified format, and the collection, archival, and access of the model output through the Earth System Grid Federation (ESGF) data replication centres have facilitated multi-model analyses.

The objective of CMIP is to better understand past, present, and future climate change arising from natural, unforced variability or in response to changes in radiative forcings in a multi-model context. Its increasing importance and scope is a tremendous success story, but this very success poses challenges for all involved. Coordination of the project has become more complex as CMIP includes more models with more processes all applied to a wider range of questions. To meet this new interest and to address a wide variety of science questions from more and more scientific research communities, reflecting the expanding scope of comprehensive modelling in climate science, has put pressure on CMIP to become larger and more extensive. Consequently, there has been an explosion in the diversity and volume of requested CMIP output from an increasing number of experiments causing challenges for CMIP's technical infrastructure (Williams et al., 2015). Cultural and organizational challenges also arise from the tension between expectations that modelling centres deliver multiple model experiments to CMIP yet at the same time advance basic research in climate science.

In response to these challenges, we have adopted a more federated structure for the sixth phase of CMIP (i.e. CMIP6) and subsequent phases. Whereas past phases of CMIP were usually described through a single overview paper, reflecting a centralized and relatively compact CMIP structure, this GMD special issue describes the new design and organization of CMIP, the suite of experiments, and its forcings, in a series of invited contributions. In this paper, we provide the overview and backdrop of the new CMIP structure as well as the main scientific foci that CMIP6 will address. We begin by describing the new organizational form for CMIP and the pressures that it was designed to alleviate (Sect. 2). It also contains a description of a small set of simulations for CMIP which are intended to be common to all participating models (Sect. 3), details of which are provided in the Appendix. We then present a brief overview of CMIP6 that serves as an introduction to the other contributions to this special issue (Sect. 4), and we close with a summary.

CMIP design – a more continuous and distributed organization

In preparing for CMIP6, the CMIP Panel (the authors of this paper), which traditionally has the responsibility for direct coordination and oversight of CMIP, initiated a 2-year process of community consultation. This consultation involved the modelling centres whose contributions form the substance of CMIP as well as communities that rely on CMIP model output for their work. Special meetings were organized to reflect on the successes of CMIP5 as well as the scientific gaps that remain or have since emerged. The consultation also sought input through a community survey, the scientific results of which are described by Stouffer et al. (2015). Four main issues related to the overall structure of CMIP were identified.

First, we identified a growing appreciation of the scientific potential to use results across different CMIP phases. Such approaches, however, require an appropriate experimental design to facilitate the identification of an ensemble of models with particular properties drawn from different phases of CMIP (e.g. Rauser et al., 2014). At the same time, it was recognized that an increasing number of Model Intercomparison Projects (MIPs) were being organized independent of CMIP, the data structure and output requirements were often inconsistent, and the relationship between the models used in the various MIPs was often difficult to determine, in which context measures to help establish continuity across MIPs or phases of CMIP would also be welcome.

Second, the scope of CMIP was taxing the resources of modelling centres making it impossible for many to consider contributing to all the proposed experiments. By providing a better basis to help modelling centres decide exactly which subset of experiments to perform, it was thought that it might be possible to minimize fragmented participation in CMIP6. A more federated experimental protocol could also encourage modelling centres to develop intercomparison studies based on their own strategic goals.

Third, some centres expressed the view that the punctuated structure of CMIP had begun to distort the model development process. Defining a protocol that allowed modelling centres to decouple their model development from the CMIP schedule would offer additional flexibility, and perhaps encourage modelling centres to finalize their models and submit some of their results sooner on their own schedule.

Fourth and finally, many groups expressed a desire for particular phases of CMIP to be more than just a collection of MIPs, but rather to reflect the strategic goals of the climate science community as, for instance, articulated by WCRP. By focusing a particular phase of CMIP around specific scientific issues, it was felt that the modelling resources could be more effectively applied to those scientific questions that had matured to a point where coordinated activities were expected to have substantial impact.

A variety of mechanisms were proposed and intensely debated to address these issues. The outcome of these discussions is embodied in the new CMIP structure, which has three major components. First, the identification of a handful of common experiments, the Diagnostic, Evaluation and Characterization of Klima (DECK) experiments (klima is Greek for “climate”), and CMIP historical simulations, which can be used to establish model characteristics and serves as its entry card for participating in one of CMIP's phases or in other MIPs organized between CMIP phases, as depicted in Fig. 1. Second, common standards, coordination, infrastructure, and documentation that facilitate the distribution of model outputs and the characterization of the model ensemble, and third, the adoption of a more federated structure, building on more autonomous CMIP-Endorsed MIPs.

Realizing the idea of a particular phase of CMIP being centred on a collection of more autonomous MIPs required the development of procedures for soliciting and evaluating MIPs in light of the scientific focus chosen for CMIP6. These procedures were developed and implemented by the CMIP Panel. The responses to the CMIP5 survey helped inform a series of workshops and resulted in a draft experiment design for CMIP6. This initial design for CMIP6 was published in early 2014 (Meehl et al., 2014) and was open for comments from the wider community until mid-September 2014. In parallel to the open review of the design, the CMIP Panel distributed an open call for proposals for MIPs in April 2014. These proposals were broadly reviewed within WCRP with the goal to encourage and enhance synergies among the different MIPs, to avoid overlapping experiments, to fill gaps, and to help ensure that the WCRP Grand Science Challenges would be addressed. Revised MIP proposals were requested and evaluated by the CMIP Panel in summer 2015. The selection of MIPs was based on the CMIP Panel's evaluation of ten endorsement criteria (Table 1). To ensure community engagement, an important criterion was that enough modelling groups (at least eight) were willing to perform all of the MIP's highest priority (Tier 1) experiments and providing all the requested diagnostics needed to answer at least one of its leading science questions. For each of the selected CMIP6-Endorsed MIPs it turned out that at least ten modelling groups indicated their intent to participate in Tier 1 experiments at least, thus attesting to the wide appeal and level of science interest from the climate modelling community.

The DECK and CMIP historical simulations

The DECK comprises four baseline experiments: (a) a historical Atmospheric Model Intercomparison Project (amip) simulation, (b) a pre-industrial control simulation (piControl or esm-piControl), (c) a simulation forced by an abrupt quadrupling of CO2 (abrupt-4×CO2) and (d) a simulation forced by a 1 % yr-1 CO2 increase (1pctCO2). CMIP also includes a historical simulation (historical or esm-hist) that spans the period of extensive instrumental temperature measurements from 1850 to the present. In naming the experiments, we distinguish between simulations with CO2 concentrations calculated and anthropogenic sources of CO2 prescribed (esm-piControl and esm-hist) and simulations with prescribed CO2 concentrations (all others). Hereafter, models that can calculate atmospheric CO2 concentration and account for the fluxes of CO2 between the atmosphere, the ocean, and biosphere are referred to as Earth System Models (ESMs).

The DECK experiments are chosen (1) to provide continuity across past and future phases of CMIP, (2) to evolve as little as possible over time, (3) to be well established, and incorporate simulations that modelling centres perform anyway as part of their own development cycle, and (4) to be relatively independent of the forcings and scientific objectives of a specific phase of CMIP. The four DECK experiments and the CMIP historical simulations are well suited for quantifying and understanding important climate change response characteristics. Modelling groups also commonly perform simulations of the historical period, but reconstructions of the external conditions imposed on historical runs (e.g. land-use changes) continue to evolve significantly, influencing the simulated climate. In order to distinguish among the historical simulations performed under different phases of CMIP, the historical simulations are labelled with the phase (e.g. “CMIP5 historical” or “CMIP6 historical”). A similar argument could be made to exclude the AMIP experiments from the DECK. However, the AMIP experiments are simpler, more routine, and the dominating role of sea surface temperatures and the focus on recent decades means that for most purposes AMIP experiments from different phases of CMIP are more likely to provide the desired continuity.

The persistence and consistency of the DECK will make it possible to track changes in performance and response characteristics over future generations of models and CMIP phases. Although the set of DECK experiments is not expected to evolve much, additional experiments may become enough well established as benchmarks (routinely run by modelling groups as they develop new model versions) so that in the future they might be migrated into the DECK. The common practice of including the DECK in model development efforts means that models can contribute to CMIP without carrying out additional computationally burdensome experiments. All of the DECK and the historical simulations were included in the core set of experiments performed under CMIP5 (Taylor et al., 2012), and all but the abrupt-4×CO2 simulation were included in even earlier CMIP phases.

Under CMIP, credentials of the participating atmosphere–ocean general circulation models (AOGCMs) and ESMs are established by performing the DECK and CMIP historical simulations, so these experiments are required from all models. Together these experiments document the mean climate and response characteristics of models. They should be run for each model configuration used in a CMIP-Endorsed MIP. A change in model configuration includes any change that might affect its simulations other than noise expected from different realizations. This would include, for example, a change in model resolution, physical processes, or atmospheric chemistry treatment. If an ESM is used in both CO2-emission-driven mode and CO2-concentration-driven mode in subsequent CMIP6-Endorsed MIPs, then both emission-driven and concentration-driven control, and historical simulations should be done and they will be identical in all forcings except the treatment of CO2.

The forcing data sets that will drive the DECK and CMIP6 historical simulations are described separately in a series of invited contributions to this special issue. These articles also include some discussion of uncertainty in the data sets. The data will be provided by the respective author teams and made publicly available through the ESGF using common metadata and formats.

The historical forcings are based as far as possible on observations and cover the period 1850–2014. These include: emissions of short-lived species and long-lived greenhouse gases (GHGs),GHG concentrations,global gridded land-use forcing data sets,solar forcing,stratospheric aerosol data set (volcanoes),AMIP sea surface temperatures (SSTs) and sea ice concentrations (SICs),for simulations with prescribed aerosols, a new approach to prescribe aerosols in terms of optical properties and fractional change in cloud droplet effective radius to provide a more consistent representation of aerosol forcing, andfor models without ozone chemistry, time-varying gridded ozone concentrations and nitrogen deposition. Some models might require additional forcing data sets (e.g. black carbon on snow or anthropogenic dust). Allowing model groups to use different forcingHere, we distinguish between an applied input perturbation (e.g. the imposed change in some model constituent, property, or boundary condition), which we refer to somewhat generically as a “forcing”, and radiative forcing, which can be precisely defined. Even if the forcings are identical, the resulting radiative forcing depends on a model's radiation scheme (among other factors) and will differ among models. data sets might better sample uncertainty, but makes it more difficult to assess the uncertainty in the response of models to the best estimate of the forcing, available to a particular CMIP phase. To avoid conflating uncertainty in the response of models to a given forcing, it is strongly preferred for models to be integrated with the same forcing in the entry card historical simulations, and for forcing uncertainty to be sampled in supplementary simulations that are proposed as part of DAMIP. In any case it is important that all forcing data sets are documented and are made available alongside the model output on the ESGF. Likewise to the extent modelling centres simplify forcings, for instance by regridding or smoothing in time or some other dimension, this should also be documented.

For the future scenarios selected by ScenarioMIP, forcings are provided by the integrated assessment model (IAM) community for the period 2015–2100 (or until 2300 for the extended simulations). For atmospheric emissions and concentrations as well as for land use, the forcings are harmonized across IAMs and scenarios using a similar procedure as in CMIP5 (van Vuuren et al., 2011). This procedure ensures consistency with historical forcing data sets and between the different forcing categories. The selection of scenarios and the main characteristics are described elsewhere in this special issue, while the underlying IAM scenarios are described in a special issue in Global Environmental Change.

An important gap identified in CMIP5, and in previous CMIP phases, was a lack of careful quantification of the radiative forcings from the different specified external forcing factors (e.g. GHGs, sulphate aerosols) in each model (Stouffer et al., 2015). This has impaired attempts to identify reasons for differences in model responses. The effective radiative forcing or ERF component of the Radiative Forcing MIP (RFMIP) includes fixed SST simulations to diagnose the forcing (RFMIP-lite), which are further detailed in the corresponding contribution to this special issue. Although not included as part of the DECK, in recognition of this deficiency in past phases of

CMIP we strongly encourage all CMIP6 modelling groups to participate in RFMIP-lite. The modest additional effort would enable the radiative forcing to be characterized for both historic and future scenarios across the model ensemble. Knowing this forcing would lead to a step change in efforts to understand the spread of model responses for CMIP6 and contribute greatly to answering one of CMIP6's science questions.

An overview of the main characteristics of the DECK and CMIP6 historical simulations appears in Table 2. Here we briefly describe these experiments. Detailed specifications for the DECK and CMIP6 historical simulations are provided in Appendix A and are summarized in Table A1.

The DECK

The AMIP and pre-industrial control simulations of the DECK provide opportunities for evaluating the atmospheric model and the coupled system, and in addition they establish a baseline for performing many of the CMIP6 experiments. Many experiments branch from, and are compared with, the pre-industrial control. Similarly, a number of diagnostic atmospheric experiments use AMIP as a control. The idealized CO2-forced experiments in the DECK (abrupt-4×CO2 and 1pctCO2), despite their simplicity, can reveal fundamental forcing and feedback response characteristics of models.

For nearly 3 decades, AMIP simulations (Gates et al., 1999) have been routinely relied on by modelling centres to help in the evaluation of the atmospheric component of their models. In AMIP simulations, the SSTs and SICs are prescribed based on observations. The idea is to analyse and evaluate the atmospheric and land components of the climate system when they are constrained by the observed ocean conditions. These simulations can help identify which model errors originate in the atmosphere, land, or their interactions, and they have proven useful in addressing a great variety of questions pertaining to recent climate changes. The AMIP simulations performed as part of the DECK cover at least the period from January 1979 to December 2014. The end date will continue to evolve as the SSTs and SICs are updated with new observations. Besides prescription of ocean conditions in these simulations, realistic forcings are imposed that should be identical to those applied in the CMIP historical simulations. Large ensembles of AMIP simulations are encouraged as they can help to improve the signal-to-noise ratio (Li et al., 2015).

The remaining three experiments in the DECK are premised on the coupling of the atmospheric and oceanic circulation. The pre-industrial control simulation (piControl or esm-piControl) is performed under conditions chosen to be representative of the period prior to the onset of large-scale industrialization, with 1850 being the reference year. Historically, the industrial revolution began in the 18th century, and in nature the climate in 1850 was not stable as it was already changing due to prior historical changes in radiative forcings. In CMIP6, however, as in earlier CMIP phases, the control simulation is an attempt to produce a stable quasi-equilibrium climate state under 1850 conditions. When discussing and analysing historical and future radiative forcings, it needs to be recognized that the radiative forcing in 1850 due to anthropogenic greenhouse gas increases alone was already around 0.25 W m-2 (Cubasch, 2013) although aerosols might have offset that to some extent. In addition, there were other pre-1850 secular changes, for example, in land use (Hurtt et al., 2011), and as a result, global net annual emissions of carbon from land use and land-use change already were responsible in 1850 for about 0.6 Pg C yr-1 (Houghton, 2010). Under the assumptions of the control simulation, however, there are no secular changes in forcing, so the concentrations and/or sources of atmospheric constituents (e.g. GHGs and emissions of short-lived species) as well as land use are held fixed, as are Earth's orbital characteristics. Because of the absence of both naturally occurring changes in forcing (e.g. volcanoes, orbital or solar changes) and human-induced changes, the control simulation can be used to study the unforced internal variability of the climate system.

An initial climate spin-up portion of a control simulation, during which the climate begins to come into balance with the forcing, is usually performed. At the end of the spin-up period, the piControl starts. The piControl serves as a baseline for experiments that branch from it. To account for the effects of any residual drift, it is required that the piControl simulation extends as far beyond the branching point as any experiment to which it will be compared. Only then can residual climate drift in an experiment be removed so that it is not misinterpreted as part of the model's forced response. The recommended minimum length for the piControl is 500 years.

The two DECK climate change experiments branch from some point in the 1850 control simulation and are designed to document basic aspects of the climate system response to greenhouse gas forcing. In the first, the CO2 concentration is immediately and abruptly quadrupled from the global annual mean 1850 value that is used in piControl. This abrupt-4×CO2 simulation has proven to be useful for characterizing the radiative forcing that arises from an increase in atmospheric CO2 as well as changes that arise indirectly due to the warming. It can also be used to estimate a model's equilibrium climate sensitivity (ECS, Gregory et al., 2004). In the second, the CO2 concentration is increased gradually at a rate of 1 % per year. This experiment has been performed in all phases of CMIP since CMIP2, and serves as a consistent and useful benchmark for analysing model transient climate response (TCR). The TCR takes into account the rate of ocean heat uptake which governs the pace of all time-evolving climate change (e.g. Murphy and Mitchell, 1995). In addition to the TCR, the 1 % CO2 integration with ESMs that include explicit representation of the carbon cycle allows the calculation of the transient climate response to cumulative carbon emissions (TCRE), defined as the transient global average surface temperature change per unit of accumulated CO2 emissions (IPCC, 2013). Despite their simplicity, these experiments provide a surprising amount of insight into the behaviour of models subject to more complex forcing (e.g. Bony et al., 2013; Geoffroy et al., 2013).

CMIP historical simulations

In addition to the DECK, CMIP requests models to simulate the historical period, defined to begin in 1850 and extend to the near present. The CMIP historical simulation and its CO2-emission-driven counterpart, esm-hist, branch from the piControl and esm-piControl, respectively (see details in Sect. A1.2). These simulations are forced, based on observations, by evolving, externally imposed forcings such as solar variability, volcanic aerosols, and changes in atmospheric composition (GHGs and aerosols) caused by human activities. The CMIP historical simulations provide rich opportunities to assess model ability to simulate climate, including variability and century timescale trends (e.g. Flato et al., 2013). These simulations can also be analysed to determine whether climate model forcing and sensitivity are consistent with the observational record, which provides opportunities to better bound the magnitude of aerosol forcing (e.g. Stevens, 2015). In addition they, along with the control run, provide the baseline simulations for performing formal detection and attribution studies (e.g. Stott et al., 2006) which help uncover the causes of forced climate change.

As with performing control simulations, models that include representation of the carbon cycle should normally perform two different CMIP historical simulations: one with prescribed CO2 concentration and the other with prescribed CO2 emissions (accounting explicitly for fossil fuel combustion). In the second, CO2 concentrations are predicted by the model. The treatment of other GHGs should be identical in both simulations. Both types of simulation are useful in evaluating how realistically the model represents the response of the carbon cycle anthropogenic CO2 emissions, but the prescribed concentration simulation enables these more complex models to be evaluated fairly against those models without representation of carbon cycle processes.

Common standards, infrastructure, and documentation

A key to the success of CMIP and one of the motivations for incorporating a wide variety of coordinated modelling activities under a single framework in a specific phase of CMIP (now CMIP6) is the desire to reduce duplication of effort, minimize operational and computational burdens, and establish common practices in producing and analysing large amounts of model output. To enable automated processing of output from dozens of different models, CMIP has led the way in encouraging adoption of data standards (governing structure and metadata) that facilitate development of software infrastructure in support of coordinated modelling activities. The ESGF has capitalized on this standardization to provide access to CMIP model output hosted by institutions around the world. As the complexity of CMIP has increased and as the potential use of model output expands beyond the research community, the evolution of the climate modelling infrastructure requires enhanced coordination. To help in this regard, the WGCM Infrastructure Panel (WIP) was set up, and is now providing guidance on requirements and establishing specifications for model output, model and simulation documentation, and archival and delivery systems for CMIP6 data. In parallel to the development of the CMIP6 experiment design, the ESGF capabilities are being further extended and improved. In CMIP5, with over 1,000 different model/experiment combinations, a first attempt was also made to capture structured metadata describing the models and the simulations themselves. Based upon the Common Information Model

(CIM, Lawrence et al., 2012), tools were provided to capture documentation of models and simulations. This effort is now continuing under the banner of the international ES-DOC activity, which establishes agreements on common Controlled Vocabularies (CVs) to describe models and simulations. Modelling groups will be required to provide documentation following a common template and adhering to the CVs. With the documentation recorded uniformly across models, researchers will, for example, be able to use web-based tools to determine differences in model versions and differences in forcing and other conditions that affect each simulation. Further details on the CMIP6 infrastructure can be found in the WIP contribution to this special issue.

A more routine benchmarking and evaluation of the models is envisaged to be a central part of CMIP6. As noted above, one purpose of the DECK and CMIP historical simulations is to provide a basis for documenting model simulation characteristics. Towards that end an infrastructure is being developed to allow analysis packages to be routinely executed whenever new model experiments are contributed to the CMIP archive at the ESGF. These efforts utilize observations served by the ESGF contributed from the obs4MIPs (Ferraro et al., 2015; Teixeira et al., 2014) and ana4MIPs projects. Examples of available tools that target routine evaluation in CMIP include the PCMDI metrics software (Gleckler et al., 2016) and the Earth System Model Evaluation Tool (ESMValTool, Eyring et al., 2016), which brings together established diagnostics such as those used in the evaluation chapter of IPCC AR5 (Flato et al., 2013). The ESMValTool also integrates other packages, such as the NCAR Climate Variability Diagnostics Package (Phillips et al., 2014), or diagnostics such as the cloud regime metric (Williams and Webb, 2009) developed by the Cloud Feedback MIP (CFMIP) community. These tools can be used to broadly and comprehensively characterize the performance of the wide variety of models and model versions that will contribute to CMIP6. This evaluation activity can, compared with CMIP5, more quickly inform users of model output, as well as the modelling centres, of the strengths and weaknesses of the simulations, including the extent to which long-standing model errors remain evident in newer models. Building such a community-based capability is not meant to replace how CMIP research is currently performed but rather to complement it. These tools can also be used to compute derived variables or indices alongside the ESGF, and their output could be provided back to the distributed ESGF archive.

CMIP6

Scientific focus of CMIP6

In addition to the DECK and CMIP historical simulations, a number of additional experiments will colour a specific phase of CMIP, now CMIP6. These experiments are likely to change from one CMIP phase to the next. To maximize the relevance and impact of CMIP6, it was decided to use the WCRP Grand Science Challenges (GCs) as the scientific backdrop of the CMIP6 experimental design. By promoting research on critical science questions for which specific gaps in knowledge have hindered progress so far, but for which new opportunities and more focused efforts raise the possibility of significant progress on the timescale of 5–10 years, these GCs constitute a main component of the WCRP strategy to accelerate progress in climate science (Brasseur and Carlson, 2015). They relate to (1) advancing understanding of the role of clouds in the general atmospheric circulation and climate sensitivity (Bony et al., 2015), (2) assessing the response of the cryosphere to a warming climate and its global consequences, (3) understanding the factors that control water availability over land (Trenberth and Asrar, 2014), (4) assessing climate extremes, what controls them, how they have changed in the past and how they might change in the future, (5) understanding and predicting regional sea level change and its coastal impacts, (6) improving near-term climate predictions, and (7) determining how biogeochemical cycles and feedback control greenhouse gas concentrations and climate change. These GCs will be using the full spectrum of observational, modelling and analytical expertise across the WCRP, and in terms of modelling most GCs will address their specific science questions through a hierarchy of numerical models of different complexities. Global coupled models obviously constitute an essential element of this hierarchy, and CMIP6 experiments will play a prominent role across all GCs by helping to answer the following three CMIP6 science questions: How does the Earth system respond to forcing? What are the origins and consequences of systematic model biases? How can we assess future climate change given internal climate variability, climate predictability, and uncertainties in scenarios?

These three questions will be at the centre of CMIP6. Science topics related specifically to CMIP6 will be addressed through a range of CMIP6-Endorsed MIPs that are organized by the respective communities and overseen by the CMIP Panel (Fig. 2). Through these different MIPs and their connection to the GCs, the goal is to fill some of the main scientific gaps of previous CMIP phases. This includes, in particular, facilitating the identification and interpretation of model systematic errors, improving the estimate of radiative forcings in past and future climate change simulations, facilitating the identification of robust climate responses to aerosol forcing during the historical period, better accounting of the impact of short-term forcing agents and land use on climate, better understanding the mechanisms of decadal climate variability, along with many other issues not addressed satisfactorily in CMIP5 (Stouffer et al., 2015). In endorsing a number of these MIPs, the CMIP Panel acted to minimize overlaps among the MIPs and to reduce the burden on modelling groups, while maximizing the scientific complementarity and synergy among the different MIPs.

The CMIP6-Endorsed MIPs

Close to 30 suggestions for CMIP6 MIPs have been received so far, of which 21 MIPs were eventually endorsed and invited to participate (Table 3). Of those not selected some were asked to work with other proposed MIPs with overlapping science goals and objectives. Of the 21 CMIP6-Endorsed MIPs, 4 are diagnostic in nature, which means that they define and analyse additional output, but do not require additional experiments. In the remaining 17 MIPs, a total of around 190 experiments have been proposed resulting in 40 000 model simulation years with around half of these in Tier 1. The CMIP6-Endorsed MIPs show broad coverage and distribution across the three CMIP6 science questions, and all are linked to the WCRP Grand Science Challenges (Fig. 3).

Each of the 21 CMIP6-Endorsed MIPs is described in a separate invited contribution to this special issue. These contributions will detail the goal of the MIP and the major scientific gaps the MIP is addressing, and will specify what is new compared to CMIP5 and previous CMIP phases. The contributions will include a description of the experimental design and scientific justification of each of the experiments for Tier 1 (and possibly beyond), and will link the experiments and analysis to the DECK and CMIP6 historical simulations. They will additionally include an analysis plan to fully justify the resources used to produce the various requested variables, and if the analysis plan is to compare model results to observations, the contribution will highlight possible model diagnostics and performance metrics specifying whether the comparison entails any particular requirement for the simulations or outputs (e.g. the use of observational simulators). In addition, possible observations and reanalysis products for model evaluation are discussed and the MIPs are encouraged to help facilitate their use by contributing them to the obs4MIPs/ana4MIPs archives at the ESGF (see Sect. 3.3). In some MIPs, additional forcings beyond those used in the DECK and CMIP6 historical simulations are required, and these are described in the respective contribution as well. A number of MIPs are developments and/or continuation of long-standing science themes. These include MIPs specifically addressing science questions related to cloud feedback and the understanding of spatial patterns of circulation and precipitation (CFMIP), carbon cycle feedback, and the understanding of changes in carbon fluxes and stores (C4MIP), detection and attribution (DAMIP) that newly includes 21st-century GHG-only simulations allowing the projected responses to GHGs and other forcings to be separated and scaled to derive observationally constrained projections, and paleoclimate (PMIP), which assesses the credibility of the model response to forcing outside the range of recent variability. These MIPs reflect the importance of key forcing and feedback processes in understanding past, present, and future climate change and have developed new experiments and science plans focused on emerging new directions that will be at the centre of the WCRP Grand Science Challenges. A few new MIPs have arisen directly from gaps in understanding in CMIP5 (Stouffer et al., 2015), for example, poor quantification of radiative forcing (RFMIP), better understanding of ocean heat uptake and sea level rise (FAFMIP), and understanding of model response to volcanic forcing (VolMIP).

Since CMIP5, other MIPs have emerged as the modelling community has developed more complex ESMs with interactive components beyond the carbon cycle. These include the consistent quantification of forcings and feedback from aerosols and atmospheric chemistry (AerChemMIP), and, for the first time in CMIP, modelling of sea level rise from land ice sheets (ISMIP6).

Some MIPs specifically target systematic biases focusing on improved understanding of the sea ice state and its atmospheric and oceanic forcing (SIMIP), the physical and biogeochemical aspects of the ocean (OMIP), land, snow and soil moisture processes (LS3MIP), and improved understanding of circulation and variability with a focus on stratosphere–troposphere coupling (DynVarMIP). With the increased emphasis in the climate science community on the need to represent and understand changes in regional circulation, systematic biases are also addressed on a more regional scale by the Global Monsoon MIP (GMMIP) and a first coordinated activity on high-resolution modelling (HighResMIP).

For the first time, future scenario experiments, previously coordinated centrally as part of the CMIP5 core experiments, will be run as an MIP ensuring clear definition and well-coordinated science questions. ScenarioMIP will run a new set of future long-term (century timescale) integrations engaging input from both the climate science and integrated assessment modelling communities. The new scenarios are based on a matrix that uses the shared socioeconomic pathways (SSPs, O'Neill et al., 2015) and forcing levels of the Representative Concentration Pathways (RCP) as axes. As a set, they span the same range as the CMIP5 RCPs (Moss et al., 2010), but fill critical gaps for intermediate forcing levels and questions, for example, on short-lived species and land use. The near-term experiments (10–30 years) are coordinated by the decadal climate prediction project (DCPP) with improvements expected, for example, from the initialization of additional components beyond the ocean and from a more detailed process understanding and evaluation of the predictions to better identify sources and limits of predictability.

Other MIPs include specific future mitigation options, e.g. the land use MIP (LUMIP) that is for the first time in CMIP isolating regional land management strategies to study how different surface types respond to climate change and direct anthropogenic modifications, or the geoengineering MIP (GeoMIP), which examines climate impacts of newly proposed radiation modification geoengineering strategies.

The diagnostic MIP CORDEX will oversee the downscaling of CMIP6 models for regional climate projections. Another historic development in our field that provides, for the first time in CMIP, an avenue for a more formal communication between the climate modelling and user community is the endorsement of the vulnerability, impacts, and adaptation and climate services advisory board (VIACS AB). This diagnostic MIP requests certain key variables of interest to the VIACS community be delivered in a timely manner to be used by climate services and in impact studies.

All MIPs define output streams in the centrally coordinated CMIP6 data request for each of their own experiments as well as the DECK and CMIP6 historical simulations (see the CMIP6 data request contribution to this special issue for details). This will ensure that the required variables are stored at the frequency and resolution required to address the specific science questions and evaluation needs of each MIP and to enable a broad characterization of the performance of the CMIP6 models.

We note that only the Tier 1 MIP experiments are overseen by the CMIP Panel, but additional experiments are proposed by the MIPs in Tiers 2 and 3. We encourage the modelling groups to participate in the full suite of experiments beyond Tier 1 to address in more depth the scientific questions posed.

The call for MIP applications for CMIP6 is still open and new proposals will be reviewed at the annual WGCM meetings. However, we point out that the additional MIPs suggested after the CMIP6 data request has been finalized will have to work with the already defined model output from the DECK and CMIP6 historical simulations, or work with the modelling group to recover additional variables from their internal archives. We also point out that some experiments proposed by CMIP6-Endorsed MIPs may not be finished until after CMIP6 ends.

Summary

CMIP6 continues the pattern of evolution and adaptation characteristic of previous phases of CMIP. To centre CMIP at the heart of activities within climate science and encourage links among activities within the World Climate Research Programme (WCRP), CMIP6 has been formulated scientifically around three specific questions, amidst the backdrop of the WCRP's seven Grand Science Challenges. To meet the increasingly broad scientific demands of the climate-science community, yet be responsive to the individual priorities and resource limitations of the modelling centres, CMIP has adopted a new, more federated organizational structure.

CMIP has now evolved from a centralized activity involving a large number of experiments to a federated activity, encompassing many individually designed MIPs. CMIP6 comprises 21 individual CMIP6-Endorsed MIPs and the DECK and CMIP6 historical simulations. Four of the 21 CMIP6-Endorsed MIPs are diagnostic in nature, meaning that they require additional output from models, but not additional simulations. The total amount of output from CMIP6 is estimated to be between 20 and 40 petabytes, depending on model resolution and the number of modelling centres ultimately participating in CMIP6. Questions addressed in the MIPs are wide ranging, from the climate of distant past to the response of turbulent cloud processes to radiative forcing, from how the terrestrial biosphere influences the uptake of CO2 to how much predictability is stored in the ocean, from how to best project near-term to long-term future climate changes while considering interdependence and differences in model performance in the CMIP6 ensemble, and from what regulates the distribution of tropospheric ozone, to the influence of land-use changes on water availability.

The last 3 years have been dedicated to conceiving and then planning what we now call CMIP6. Starting in 2016, the first modelling centres are expected to begin performing the DECK and uploading output on the ESGF. Forcings for the DECK and CMIP6 historical simulations will be ready before mid-2016 so that these experiements can be started, and by the end of 2016 the diverse forcings for different scenarios of future human activity will become available. Past experience suggests that most centres will complete their CMIP simulations within a few years while the analysis of CMIP6 results will likely go on for a decade or more (Fig. 4).

Through an intensified effort to align CMIP with specific scientific questions and the WCRP Grand Science Challenges, we expect CMIP6 to continue CMIP's tradition of major scientific advances. CMIP6 simulations and scientific achievements are expected to support the IPCC Sixth Assessment Report (AR6) as well as other national and international climate assessments or special reports. Ultimately scientific progress on the most pressing problems of climate variability and change will be the best measure of the success of CMIP6.

Data availability

The model output from the DECK and CMIP6 historical simulations described in this paper will be distributed through the Earth System Grid Federation (ESGF) with digital object identifiers (DOIs) assigned. As in CMIP5, the model output will be freely accessible through data portals after registration. In order to document CMIP6's scientific impact and enable ongoing support of CMIP, users are obligated to acknowledge CMIP6, the participating modelling groups, and the ESGF centres (see details on the CMIP Panel website at http://www.wcrp-climate.org/index.php/wgcm-cmip/about-cmip). Further information about the infrastructure supporting CMIP6, the metadata describing the model output, and the terms governing its use are provided by the WGCM Infrastructure Panel (WIP) in their invited contribution to this special issue. Along with the data, the provenance of the data will be recorded, and DOIs will be assigned to collections of output so that they can be appropriately cited. This information will be made readily available so that published research results can be verified and credit can be given to the modelling groups providing the data. The WIP is coordinating and encouraging the development of the infrastructure needed to archive and deliver this information. In order to run the experiments, data sets for natural and anthropogenic forcings are required. These forcing data sets are described in separate invited contributions to this special issue. The forcing data sets will be made available through the ESGF with version control and DOIs assigned.